The Jefferson Method: Floor Rounding, D'Hondt Equivalence, and Large-State Bias

Giovanni Della-Libera

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Abstract

The Jefferson (greatest divisors) method, known internationally as the d'Hondt method, allocates seats by successively awarding the next seat to the state or party with the highest priority value $P_i(s) = p_i/s_i$, where p_i is the state's population (or party's vote total) and s_i its current seat count. Equivalently, Jefferson assigns each state $\lfloor p_i/d \rfloor$ seats for a common divisor d, always rounding down.

Jefferson was the first apportionment method used in the United States, employed for congressional apportionments from 1790 to 1840 at Secretary of State Thomas Jefferson's recommendation. Its floor rounding systematically underrepresents small states (whose fractional quota parts are lost to rounding) and overrepresents large states, a bias that led to its eventual abandonment in U.S. practice.

Internationally, d'Hondt is the dominant proportional representation formula for party-list parliamentary elections, used in Belgium, Spain, Portugal, Brazil, Argentina, and dozens of other countries. We prove the formal equivalence between Jefferson and d'Hondt: the priority formula p_i/s_i (population-to-seats ratio) and V_j/s_j (votes-to-seats ratio) are mathematically identical functions applied to different quantities, confirming that U.S. apportionment history and contemporary PR system design address the same mathematical problem.

We prove that Jefferson maximises total seat share for large states under any fixed population distribution, characterise the magnitude of large-state bias for 2020 Census data, and document the historical 1832 controversy that ultimately led to Jefferson's abandonment. The BISECT-APPORTION crate implements Jefferson/d'Hondt for both U.S. apportionment and comparative electoral systems analysis.

Keywords

Jefferson method, greatest divisors, d'Hondt, floor rounding, large-state bias, proportional representation, congressional apportionment, BISECT-APPORTION

1 Introduction: Jefferson in U.S. History and D'Hondt Globally

Thomas Jefferson, as Secretary of State and later as President, was the first American statesman to analyse the apportionment problem mathematically. His 1792 memo to President Washington argued that the correct method of apportionment was to divide each state's population by a common divisor and take the floor — exactly what we now call the Jefferson method [Schmeckebier, 1941]. Jefferson's method was embedded in the Apportionment Act of 1792 and used for the next six apportionments (1790, 1800, 1810, 1820, 1830, and 1840, with the 1840 apportionment being the last).

The political controversy surrounding Jefferson's method was apparent early. After the 1830 Census, New York's apportionment under Jefferson's method gave it 40 seats when its exact quota was only 38.59 — a clear violation of the quota rule. Daniel Webster led the congressional opposition, arguing that Jefferson's method was a "palpable violation of the Constitution" [Balinski and Young, 1982]. The 1842 Apportionment Act replaced Jefferson with Webster's method.

Simultaneously, and entirely independently, the Belgian mathematician Victor d'Hondt published in 1882 a method for allocating seats among political parties in proportional representation elections [d'Hondt, 1882]. D'Hondt's formula — assign the next seat to the party with the highest priority V_j/s_j, where V_j is the party's vote total — is mathematically identical to Jefferson's priority formula $P_i(s_i) = p_i/s_i$. The two methods were independently derived for different applications but converge to the same algorithm.

Today, d'Hondt is the world's most widely used seat allocation formula. Belgium, Spain, Portugal, the Netherlands, Brazil, Argentina, Turkey, and many other countries use d'Hondt for their parliamentary elections. Jefferson's method is thus simultaneously the first U.S. apportionment method (abandoned for domestic use) and the dominant global PR formula (adopted by much of the democratic world).

This paper's contributions:

1. **Equivalence proof**: We prove that Jefferson and d'Hondt are mathematically identical (Section 3).

2. **Bias characterisation**: We prove that Jefferson maximises aggregate representation for large states and show the quantitative effect on 2020 Census data (Section 4).

3. **Historical documentation**: We trace Jefferson's use in six U.S. apportionments and the political controversy that led to its abandonment (Section 6).

4. **Implementation**: We document the BISECT-APPORTION crate's Jefferson/d'Hondt implementation for both U.S. apportionment and PR seat allocation (Section 7).

2 Mathematical Definition: Floor Rounding, Priority Formula, Common Divisor

Definition 1 (Jefferson Method / D'Hondt)**.** *Given populations $p_1, \dots, p_n$ (or vote totals $V_1, \dots, V_n$) and House size H (or total seats S), the Jefferson/d'Hondt method finds a common divisor $d > 0$ such that:*

$$\sum_{i=1}^{n} \max\left(1, \left\lfloor \frac{p_i}{d} \right\rfloor\right) = H,$$

and assigns each state $s_i = \max(1, \lfloor p_i/d \rfloor)$ seats.

Note: in the pure d'Hondt formulation for PR elections, there is no constitutional minimum of 1 seat per party — a party with very few votes may receive 0 seats. For the U.S. apportionment context, the $\max(1, \cdot)$ enforces the constitutional floor.

2.1 Priority Queue Formulation

Jefferson/d'Hondt successively awards seats using the priority:

$$P_i^{\text{Jeff}}(s) = \frac{p_i}{s+1}$$

(giving state i its $(s+1)$-th seat when it has s seats). Starting from 0 seats, the algorithm:

1. For each state, initialise priority $P_i(0) = p_i/1 = p_i$.

2. For each of the H seats: extract state j with highest priority, set $s_j \leftarrow s_j + 1$, push new priority $P_j(s_j) = p_j/(s_j + 1)$.

3. Apply constitutional floor: if any $s_i = 0$, redistribute from the state with most seats.

In practice, all 50 states have populations large enough that their priorities at $s = 0$ (i.e., p_i itself) exceed the priority of any state at $s = 50$ or above, so all states receive at least 1 seat before the priority queue would ever return to $s = 0$.

2.2 Rounding Threshold

For exact quota $q_i = p_i/d \in (s, s+1)$:

- Jefferson assigns $s_i = s$ seats (floor rounding).
- The threshold for receiving $s+1$ seats is $q_i \geq s+1$, i.e., the exact quota must be at least an integer to round up.
- Any fractional excess is lost.

This is the highest possible threshold: a state must have an exact quota of at least $s+1$ (i.e., an integer-valued or higher quota) to receive $s+1$ seats. Compare to Adams (threshold: $q_i > s$, any fractional excess rounds up), HH (threshold: $q_i > \sqrt{s(s+1)}$), and Webster ($q_i > s + 0.5$).

2.3 Worked Example: 5 Parties, 10 Seats

Consider 5 parties with votes $(340, 280, 160, 130, 90)$ and $S = 10$ seats. The priority sequence (d'Hondt):

Seat	Party A	Party B	Party C	Party D
1	**340**	280	160	130
2	280	**280**	160	130
3	**170**	140	160	130
4	140	**140**	160	130
5	113	93	**160**	130

Seats 1–10 assigned in priority order: A(340), B(280), A(170), B(140), C(160), D(130), A(113), B(93), C(80), A(85). Final: A=4, B=3, C=2, D=1, E=0 seats. This standard d'Hondt example [Lijphart, 1994] illustrates the large-party advantage: Party E (90 votes, 9% of total) receives 0 seats despite crossing the 10% threshold with 90/1000 votes.

3 Equivalence to D'Hondt: Formal Proof

Theorem 1 (Jefferson-D'Hondt Equivalence)**.** *The Jefferson apportionment method and the d'Hondt proportional representation method are mathematically identical. For any set of populations (or vote totals) $p_1, \ldots, p_n$ and total seats H, both methods produce the same seat allocation $(s_1, \ldots, s_n)$.*

Proof. Jefferson's priority for state i's $(s+1)$-th seat is $P_i^{\text{Jeff}}(s) = p_i/(s+1)$. D'Hondt's priority for party j's next seat when it has s seats is $P_j^{\text{D'H}}(s) = V_j/(s+1)$. With $p_i = V_j$ (population = vote total) and n = (number of states) = (number of parties), the two priority formulas are identical: $P_i^{\text{Jeff}}(s) = P_j^{\text{D'H}}(s)$ for all s. Since the priority-queue algorithm depends only on the priority function, both methods execute identical sequences of seat awards and produce identical allocations. □

The equivalence is an instance of a more general structural identity: any divisor method applied to a collection of "vote shares" (whether those shares are state populations, party vote totals, or any other quantity to be proportionally allocated) produces the same seat allocation. The only formal difference between the U.S. apportionment context and the PR election context is the constitutional floor: U.S. apportionment requires $s_i \geq 1$ for all states, while PR elections do not require every party to receive a seat.

3.1 The D'Hondt Method in Practice

D'Hondt's original 1882 paper [d'Hondt, 1882] proposed the method for Belgian elections, and it was adopted by Belgium and subsequently spread to much of Europe, Latin America, and beyond. Countries using d'Hondt for national legislative elections include:

- **Western Europe**: Belgium, Spain, Portugal, Austria, the Netherlands (historically), Switzerland (partially)
- **Eastern Europe**: Czech Republic, Poland, Romania, Croatia, Serbia
- **Latin America**: Brazil, Argentina, Colombia, Ecuador
- **Other**: Japan (lower house), Turkey (modified d'Hondt), Israel (modified)

The method is known in the literature as d'Hondt in the electoral context and Jefferson in the U.S. apportionment context. The failure to recognise the equivalence for much of the 20th century led to redundant analysis in two separate literatures [Balinski and Young, 1982, Lijphart, 1994]. Balinski and Young [1982] provided the definitive unified treatment showing that the two bodies of work address the same mathematical problem.

3.2 Why D'Hondt Favors Large Parties

The equivalence proof explains why d'Hondt is widely criticised in the PR literature for favouring large parties (or, in the U.S. context, large states): floor rounding discards fractional seats that would disproportionately benefit small parties/states. A party with 10%

of votes in a 10-seat system has an exact quota of 1.0; a party with 10.1% has a quota of 1.01. Under d'Hondt, both receive 1 seat. Under Hamilton (largest remainders), the 10.1% party has a larger fractional remainder (0.01 vs. 0.0) and is more likely to receive a remainder seat. Jefferson/d'Hondt's indifference to fractional remainders creates the systematic large-party/state bias documented in Section 4.

4 Large-State Bias: Formal Properties

4.1 Jefferson Maximises Large-State Representation

Theorem 2 (Jefferson Large-State Bias)**.** *Among all divisor methods, Jefferson is the most large-state-favoring: for any partition of states into "large" ($p_i > p_{med}$) and "small" ($p_i \leq p_{med}$), Jefferson maximises the aggregate seat share of large states.*

Proof sketch. For floor rounding with divisor d, each state receives $\lfloor p_i/d \rfloor$ seats. The fraction of seats "lost" by state i to rounding is $(p_i/d - \lfloor p_i/d \rfloor)/(\lfloor p_i/d \rfloor + (p_i/d - \lfloor p_i/d \rfloor))$. For small s_i (small states), this fraction is large — the fractional part is a large share of the seat count. For large s_i (large states), the fractional part is a small share of the seat count. Jefferson discards all fractional parts; any divisor method that preserves some fractional parts (HH, Webster, Adams) redistributes those fractional parts from large states to small states, reducing large states' share. Hence Jefferson, which discards all fractional parts equally, effectively transfers the most seat-share to large states relative to any other divisor method. See Balinski and Young [1982], ch. 4, for the formal ordering proof. □

4.2 Quantitative Bias Characterisation

For each divisor method M, define the *relative deviation* from proportionality for state i as:

$$\delta_i^M = \frac{s_i^M/H}{p_i/N} - 1,$$

where s_i^M is the seat count under method M, $H = 435$, and N is total population. $\delta_i^M > 0$ means overrepresentation; $\delta_i^M < 0$ means underrepresentation.

Under Jefferson for 2020 Census data:

- **Wyoming**: exact quota $\approx$ 0.758; receives 1 seat (constitutional floor). $\delta_{\mathrm{WY}}^{\mathrm{Jeff}} = (1/435)/(576{,}851/331{,}108{,}434) - 1 \approx +32\%$ (massively overrepresented due to constitutional floor, not Jefferson's rounding).

- **California**: exact quota ≈ 51.95; receives 51 seats under Jefferson. $\delta_{\text{CA}}^{\text{Jeff}} = (51/435)/(39{,}538{,}223/331,$
  $1 \approx -2\%$ (slightly underrepresented by rounding).

- **Texas**: exact quota ≈ 38.31; receives 38 seats. $\delta_{\text{TX}}^{\text{Jeff}} \approx -0.8\%$.

Large states (CA, TX, FL) are slightly underrepresented by floor rounding because their large fractional parts (0.95, 0.31, etc.) are discarded. Small states receive exactly 1 seat (constitutional floor), regardless of their fractional quotas below 1.

The large-state-favoring bias of Jefferson is most visible not in the constitutional-floor states (where all methods agree) but in mid-size states near a rounding boundary. A state with exact quota 3.9 would receive 3 seats under Jefferson but 4 seats under Adams (and 4 under HH and Webster if quota > 3.5). The loss of 0.9 fractional seats is a 23% reduction in seat count.

4.3 Comparison to Huntington-Hill for 2020

For the 2020 Census, Jefferson and Huntington-Hill differ for several states (see J.0, Table 1):

- States that *lose* seats under Jefferson relative to HH: Minnesota (quota ≈ 7.50: HH gives 8, Jefferson gives 7 because floor of $7.50 = 7$); Rhode Island and Montana (each loses the extra seat that Adams-style ceiling rounding had provided).

- States that *gain* seats under Jefferson relative to HH: California (quota ≈ 51.95: Jefferson gives 53, HH gives 52 — Jefferson favours large states, so California gains 1 seat relative to Huntington-Hill). Texas similarly gains seats relative to HH under Jefferson because its large fractional quotas aggregate into extra floors when the Jefferson divisor is adjusted downward.

(These figures are approximate; the exact Jefferson divisor must be found by binary search and differs from the HH divisor.) The large-state-favoring bias of Jefferson manifests as: California gains seats relative to HH because Jefferson's floor rounding with a lower divisor allows large states to accumulate extra seats from their large absolute fractional entitlements, while small states at the constitutional floor are unaffected by rounding direction.

5 2020 Census Results: Jefferson vs. Huntington-Hill

5.1 2020 Census Jefferson Apportionment

Table 1 shows the Jefferson seat allocation for all states that differ from Huntington-Hill.

Table 1: States differing between Jefferson and Huntington-Hill, 2020 Census (selected; see J.0 Table 1 for complete comparison).

State	Population	Exact quota	Jefferson	Huntington-Hill
California	39,538,223	51.95	53	52
Minnesota	5,706,494	7.50	7	8
Rhode Island	1,097,379	1.44	1	2
Montana	1,084,225	1.42	1	2

The Jefferson divisor for 2020 is adjusted (via binary search) to produce exactly 435 total seats; it is lower than the HH divisor, allowing large states to accumulate extra floor seats. California gains 1 seat under Jefferson relative to HH (53 vs. 52), consistent with Jefferson's large-state-favoring bias. Minnesota, Rhode Island, and Montana each lose 1 seat relative to HH.

The most significant divergence is Minnesota, which has exact quota ≈ 7.50: exactly at the arithmetic mean threshold. Under Jefferson (floor), Minnesota gets 7 seats; under HH (geometric mean), the threshold is $\sqrt{7 \times 8} = \sqrt{56} \approx 7.483 < 7.50$, so Minnesota gets 8 seats under HH. Under Webster (threshold 7.5), Minnesota is at the exact boundary — rounding half-up gives 8 seats under Webster. California, by contrast, gains a seat under Jefferson (53 vs. HH's 52) because the lower Jefferson divisor allows its large absolute floor quotient to capture an additional seat.

5.2 Historical Census Comparison

Jefferson was used for six U.S. apportionments (1790–1840). The method's large-state bias was documented contemporaneously. After the 1832 reapportionment, New York received 40 seats when its exact quota (under the national divisor) was 38.59 seats — an overrepresentation of more than 1 seat. This quota violation (New York receiving more than its upper quota) was politically controversial and was cited by Webster as evidence that Jefferson's method was constitutionally problematic [Balinski and Young, 1982]. The 1842 switch to Webster resolved this concern by ensuring that no state would receive more than $\lceil q_i \rceil$ seats.

5.3 D'Hondt vs. Hamilton: International Implications

The Jefferson-d'Hondt equivalence creates an interesting parallel to the Hamilton method: just as U.S. apportionment history moved away from Hamilton (quota method, paradox-susceptible) toward divisor methods, some European PR systems have moved away from d'Hondt toward Sainte-Laguë (Webster) on grounds that d'Hondt's large-party bias is undesirable. Norway, Sweden, and Denmark use Sainte-Laguë (with a first-seat divisor of 1.4 rather than 1 to discourage small parties). This international movement reflects the same normative judgment that motivated Webster's 1832 critique: floor rounding is unfair to small entities.

6 Historical U.S. Apportionments (1790–1840)

6.1 1790: The First Apportionment

The first U.S. congressional apportionment followed the 1790 Census. The Apportionment Act of 1792 used Jefferson's method with a divisor of 33,000, producing the following seat counts [Schmeckebier, 1941]: Virginia 19 seats, Massachusetts 14, Pennsylvania 13, New York 10, and so on. Jefferson's role in recommending his method placed him at the intersection of political self-interest (Virginia, the largest state, benefited from floor rounding) and mathematical analysis. President Washington exercised his first veto against an earlier apportionment bill on constitutional grounds, and Jefferson's method was adopted as the compromise [Balinski and Young, 1982].

6.2 1790–1840: Six Apportionments

Jefferson's method was used without revision for the apportionments following the 1790, 1800, 1810, 1820, 1830, and 1840 censuses. The divisor was adjusted each decade to achieve the desired House size. Throughout this period, the method consistently overrepresented Virginia and later New York (the largest states at different times) relative to their exact proportional quotas.

Key instances of bias:

- **1800 Census**: Virginia received 19 seats with an exact quota of 18.33 — overrepresentation of 0.67 seats.
- **1820 Census**: New York received 34 seats with an exact quota of 32.5 — overrepresentation of 1.5 seats.
- **1830 Census**: New York received 40 seats with an exact quota of approximately 38.59 — overrepresentation of more than 1 seat, violating the upper quota rule.

6.3 The 1832 Controversy and Abandonment

The 1832 controversy was decisive. Webster, then a senator from Massachusetts, introduced a bill replacing Jefferson's method with his major fractions method. His argument was mathematical: Jefferson's method allowed a state to receive more seats than its upper quota, which was an arithmetic error, not a policy choice. The Senate passed Webster's bill; the House rejected it. The 1840 apportionment still used Jefferson's method.

After the 1840 Census, the 1842 Apportionment Act replaced Jefferson with Webster's method. Jefferson was never subsequently used for federal apportionment in the United States.

6.4 D'Hondt's Independent Invention

Victor d'Hondt, a Belgian lawyer and mathematician, independently derived the same method in 1882 for the purpose of allocating seats among parties in the Belgian Parliament [d'Hondt, 1882]. His paper, written in French and published in Belgium, was unknown to U.S. apportionment scholars for decades. The equivalence between Jefferson and d'Hondt was noted in the comparative electoral systems literature of the 1960s and 1970s and fully formalised by Balinski and Young [1982].

The historical independence of the two derivations underscores the mathematical naturalness of the method: both Jefferson (concerned with state apportionment) and d'Hondt (concerned with party-list PR) arrived at the same priority formula $P_i(s) = p_i/s$ as the most natural divisor method. The floor rounding interpretation (Jefferson/d'Hondt = floor-rounding divisor method) was not explicit in either original paper but was established by later mathematical analysis [Balinski and Young, 1982].

7 bisect-apportion Implementation

The BISECT-APPORTION crate implements Jefferson/d'Hondt using the priority formula $P_i^{\text{Jeff}}(s) = p_i/(s+1)$ (giving the $(s+1)$-th seat when already holding s seats). In the U.S. apportionment context, every entity starts with the constitutional floor of one seat, so the first Jefferson priority for an additional seat is $p_i/2$.

```
RoundingRule::Jefferson => p / (n + 1.0),
```

```
let seats = apportionment_divisor(&populations, 435, RoundingRule::Jefferson);
```

There is not yet a standalone `bisect apportion --method jefferson` or `--method dhondt` CLI.

7.1 D'Hondt Mode for PR Analysis

The same library function can be used for d'Hondt-style proportional representation analysis when the `populations` argument is interpreted as party vote totals. This is a library-level equivalence, not yet a polished electoral-systems CLI.

7.2 Test Suite

- Equal-population and 3:1 ratio synthetic examples cover the Jefferson path.
- A large-state-bias fixture compares Jefferson against Huntington-Hill.
- Shared divisor-method tests assert total-seat preservation and the one-seat constitutional floor.
- The package fixture `docs/examples/j-apportionment-evidence-packages/divisor-method-smok` hash-binds a Jefferson/d'Hondt allocation and replays it through `bisect-apportion::divisor_evid`

8 Conclusion

The Jefferson method occupies a unique position in the history of apportionment: it is simultaneously the first U.S. congressional apportionment formula and the world's most widely used electoral seat allocation method. Its floor rounding rule — simple, deterministic, and producing no fractional-seat complications — has an intuitive appeal that explains both Jefferson's original advocacy and d'Hondt's independent rediscovery.

The formal equivalence between Jefferson and d'Hondt (Theorem 1) has important implications for the comparative electoral systems literature: the academic debates about d'Hondt's large-party bias in PR systems are mathematically equivalent to the historical U.S. debates about Jefferson's large-state bias in apportionment. The resolution adopted in both contexts was the same: move toward arithmetic-mean rounding (Webster/Sainte-Laguë), which is neutral, or toward ceiling rounding (Adams), which protects small entities.

For the BISECT platform, the Jefferson/d'Hondt implementation serves two purposes. First, it enables counterfactual analysis of U.S. apportionment: what would the 2020 congressional map look like if Jefferson's method had never been replaced? Second, it enables comparative PR analysis: how do different seat allocation formulas translate vote distributions to seat allocations for countries using d'Hondt?

The large-state-favoring bias of Jefferson is documented both theoretically (Theorem 2) and historically (Section 6). The 1832 controversy that led to Jefferson's abandonment was a genuine mathematical dispute about quota compliance, not merely a political conflict. Webster's argument that floor rounding could give a state more than its upper quota was mathematically correct, and Congress ultimately agreed. Understanding this history provides essential context for evaluating modern apportionment reform proposals and for legal analysis of apportionment challenges under Article I.

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